Harshit Yadav

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Education

Indian Institute of Technology (IIT), Kharagpur 2018 – 2023
Integrated Bachelor and Master of Science, Department of Mathematics and Computing **CGPA: 8/10**

All India Senior School Certificate Examination (AISSCE, Class 12) 2018
SR Public Senior Secondary School, CBSE **Aggregate: 94.3%**

All India Senior School Certificate Examination (AISSCE, Class 10) 2016
Atomic Energy Central School, CBSE **CGPA: 10/10**

Industry Experience

Software Development Engineer - 2, NAVI Technologies, Bangalore Sept 2024 – Present
Domain: High-Performance ML Systems, LLMs, Distributed Computing; ***Instructor in Fresher BootCamp***

Embedding Generator:

- Built an autonomous AI agent that researches ML embedding model architectures from the web and automatically translates them into ready-to-run code based on user's inputs like data-source and use-case.
- Designed the system flow to connect a **Knowledge Miner**, to find and filter high-quality research papers directly to a **Pipeline Generator**, using LLMs and templates to write the PyTorch training scripts.
- Led the system design, accelerating the research cycle by **85%,** and scaling the volume of tested architectures by 12x. Mentored a junior engineer on setting up the agent's observability using LangSmith.

Model Inference Service:

- Developed a performance optimization inference solution with a **Cython**-optimized package supporting business logic for **100+** version-enabled models on GPUs, achieving **10ms** latency and **100k** RPM.
- Reduced AWS costs by **60%** by dynamic batching, async gRPC calls via **Triton** Inference server.
- Established an end-to-end multi-version model scoring ecosystem on Databricks, deploying the Cython package for offline batch processing and the Triton server via Databricks Container Service.

Oncall Automation:

- Architected a multi-agent with **LangGraph**, investigates production alerts by extracting and summarizing logs from K8s, Prometheus etc, cross-referencing them against historical incidents, suggest immediate fixes.
- Processed **100+** weekly alerts and reducing MTTR by 90% and saving engineers weekly bandwidth.

Data Aggregator Service:

- Engineered a high-performance, **distributed** Kotlin data aggregator, for accessibility to ML/DL Models.
- Developed a robust data integration layer using topological sort to consolidate diverse sources.
- Streamlined data pipelines utilizing **gRPC**, **GraphQL**, and S3, Parquet transformation with **Airflow**.

Software Development Engineer - 1, NAVI Technologies, Bangalore July 2023 – Aug 2024
Domain: Software Development; ***Awarded Navi Spearhead out of 40 DS team members.***

Model Inference Orchestrator:

- Architected model serving in **Golang**, achieving an **80% AWS** cost reduction and **95% latency** drop.
- Implemented robust model metadata management with Postgres, Redis, and Gocache for cost reduction.
- Built data pipelines with S3 and **Kafka** for data archival and used Prometheus for proactive monitoring.
- Developed a self-service model onboarding platform, enhancing deployment efficiency and model reliability.

Software Developer Intern, Samsung Research Institute, Bangalore May 2022 – June 2022
Domain: Software Development, Natural Language Understanding, Machine Learning; **[REPORT]**

- Developed a Hindi Chatbot for Samsung Finance using RASA with the NLU training datasets, domains.
- Created workflows for multilingual support via Rasa NLU servers.**Received PPO, mentored intern.**

Full Stack Developer Intern, Leoforce Technologies May 2019 – Jun 2019
Domain: JavaScript, HTML, SQL; **[CERTIFICATE]**

- Developed a management system for the employees to gather feedback with various dashboards and MBOs.

Projects

SymPy; Jan 21 – April 21
Domain: Python, Algorithms; **[PROPOSAL]**, **[CONTRIBUTIONS]**

- Architected the **DomainMatrix** module, optimizing matrix operations for increased efficiency.
- Functionality enhancements for SymPy modules like Matrix, Core, Polys, Derivatives, and Integrals.

Deep Researcher: Cost-optimized, Iterative Web Research System with Confidential Data Handling

- Architected a multi-agent research system using **LangChain** with a **supervisor-sub-agent** architecture, achieving a **30x** cost reduction through parallelized task delegation to **Minions**.
- Engineered a hybrid **RAG** strategy combining the **Tavily** API for web research with a secure **Qdrant** vector store for semantic search on confidential data with custom Function calling.
- Reduced hallucination by **38%** by enforcing a JSON schema and generating reports with citations.

LLM-based UW: LLM Platform for Underwriting Automation & Customer Re-Verification

- Developed a multi-agent underwriting platform in **LangGraph** with a meta-LLM aggregator and a **Critic Agent** pattern, increasing monthly loan disbursals by **10-15Cr** and improving approvals by **36 bps**.
- Built a proprietary document intelligence service using a **RAG** pipeline with **Qdrant**, cutting vendor costs by **₹10-15 lakhs** monthly by enabling semantic search for citable evidence.
- Implemented **Context Quarantine** to isolate the customer re-verification AI, preventing data leakage and ensuring focused analysis in a multi-component system.

Technical Skills

- **Programming Languages:** Go, Python, Cython, Kotlin, C/C++, MySQL, HTML, JavaScript
- **Frameworks & Tools:** SpringBoot, Docker, FastAPI, Ray, BentoML, RASA, PostgreSQL, REST, Gin, Locust, S3, Redis, K8, GOCD, Git, Kafka, Azure AI Services, LangGraph, LangChain, LangSmith, Qdrant
- **Libraries:** C++ STL, SymPy, NumPy, Pandas, Numba

Other Awards and Achievements

- Acquired a top **1.7%** rank in IIT-JEE Advanced-2018 and a top **0.4%** rank in IIT-JEE Mains-2018.
- Secured **3rd** position in Kshitij, IIT Kharagpur, a simulation-based coding event for soccer-playing bots.
- Awarded **INSPIRE** scholarship for Innovation in Science Pursuit at IIT Kharagpur.
- Member of Mathematics, Chess, and Basketball Team for MMM hall in General Championship.